

# Multimedia Computing

By

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# What is sound ?

- ▶ Sound is a wave phenomenon like light, but it is macroscopic and involves molecules of air being compressed and expanded under the action of some physical device.
- ▶ For example, a speaker in an audio system vibrates back and forth and produces a longitudinal pressure wave that we perceive as sound



Without air there is no sound— example: in space

# What is sound ?

- ▶ Since sound is a pressure wave, it takes on continuous values, as opposed to digitized ones with a finite range.
- ▶ Although such pressure waves are longitudinal, they still have ordinary wave properties and behaviors, such as **reflection (bouncing)**, **refraction (change of angle when entering a medium with a different density)**, and **diffraction (bending around an obstacle)**. This makes the design of “surround sound” possible.
- ▶ Since sound consists of measurable pressures at any 3D point, we can detect it by measuring the pressure level at a location, using a transducer to convert pressure to voltage levels

# Digital data acquisition

**Multimedia systems involve three major components:**

- a) Multimedia content creation
- b) Compression/storage of multimedia content, and
- c) Delivery or distribution of multimedia content

# Digital data acquisition

- ▶ Creating multimedia content using text, audio, video, and images is to record these individual media types into a digital form, making it is easy to combine and assemble these heterogeneous entities.
- ▶ Digitizing one-dimensional (such as audio)
- ▶ Two-dimensional (such as images)
- ▶ Three-dimensional (such as video)

# ANALOG AND DIGITAL SIGNALS

- ▶ A signal is analog if it can be represented by a continuous function. For instance, it might encode the changing amplitude with respect to an input dimension(s).
- ▶ Digital signals, on the other hand, are represented by a discrete set of values defined at specific (and most often regular) instances of the input domain, which might be time, space, or both

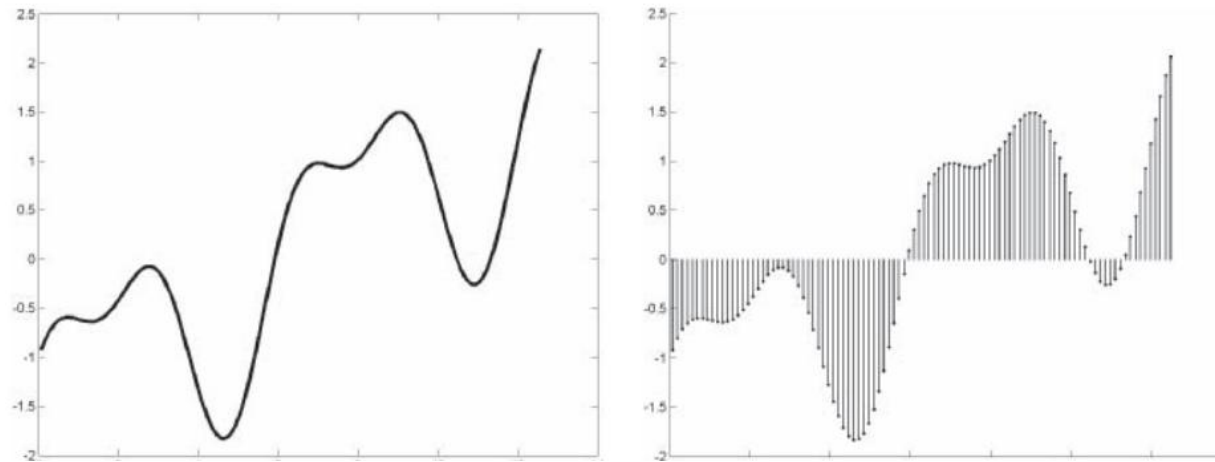
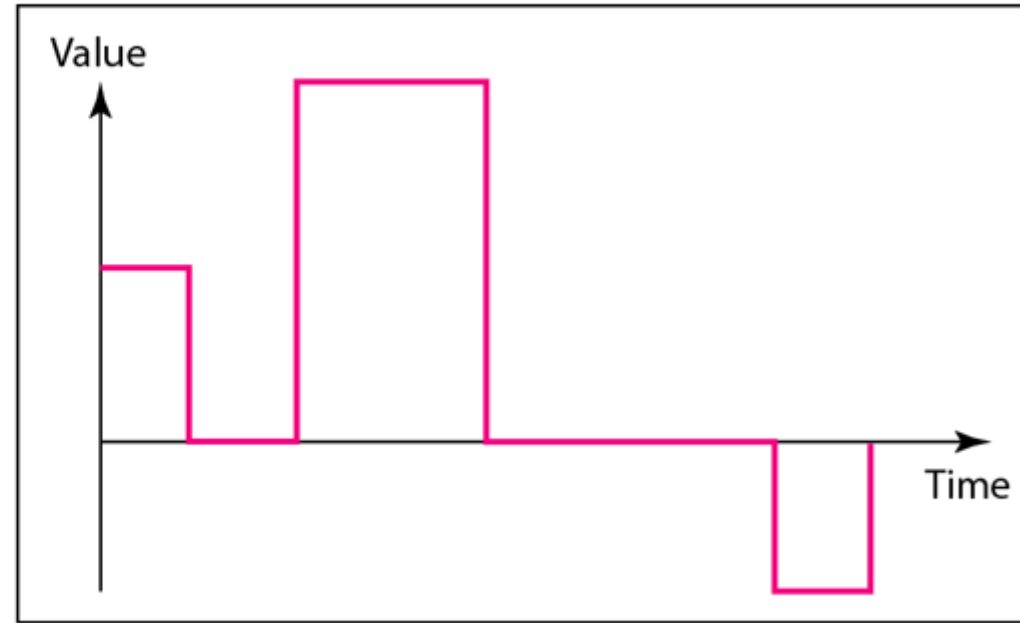


Figure 2-1 Example of an analog signal (left) and a digital signal (right) in one dimension

# ANALOG AND DIGITAL SIGNALS



a. Analog signal



b. Digital signal

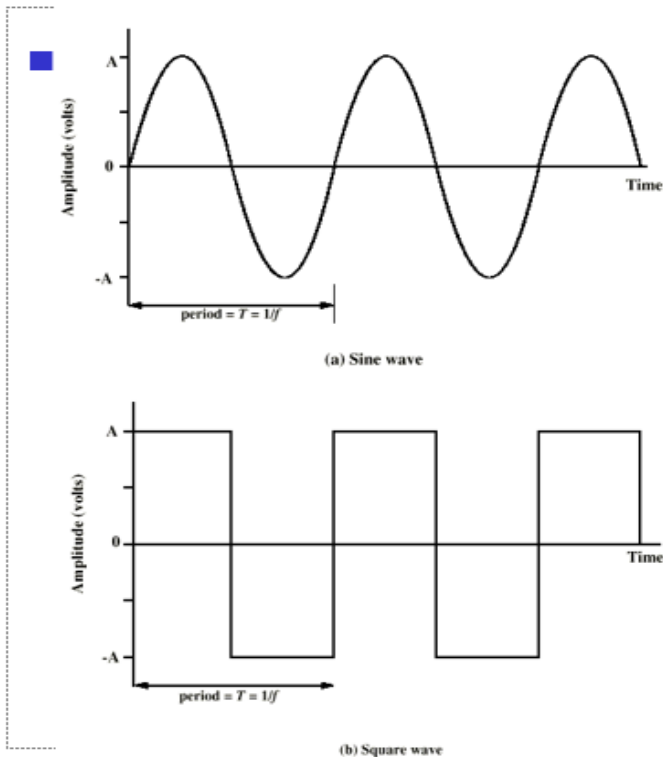


# Advantages of digital signals over analog

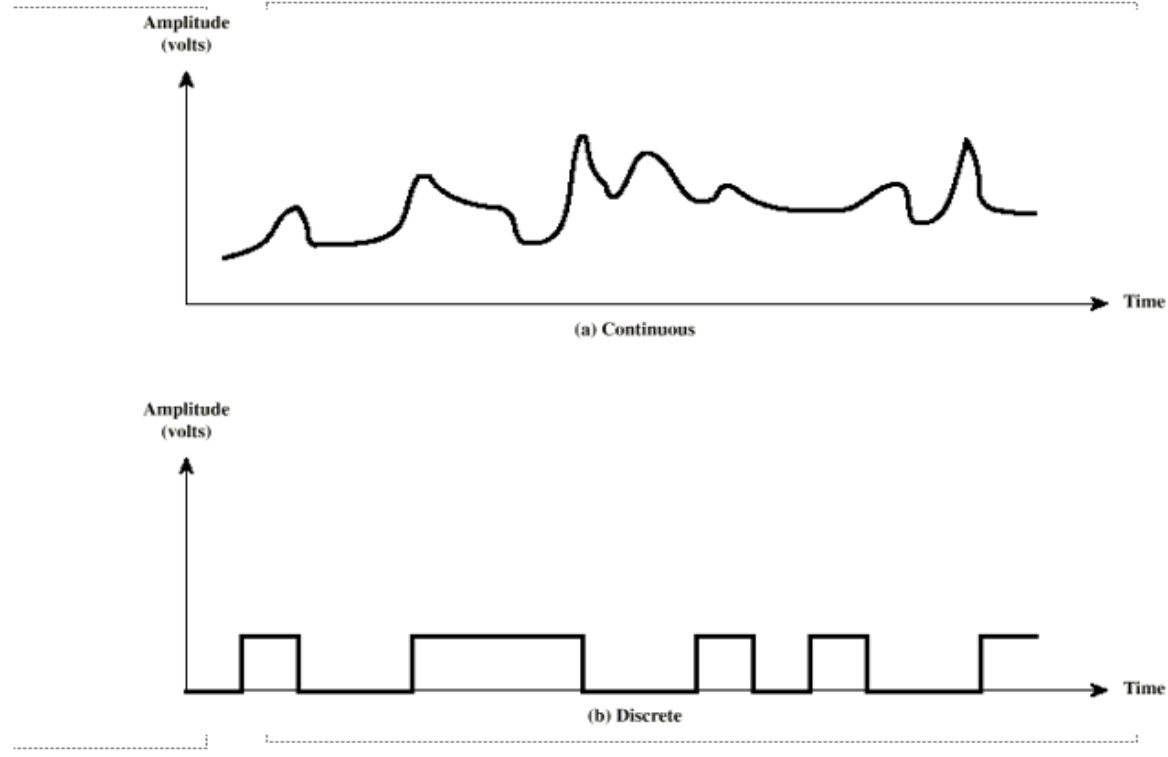
- ▶ When media is represented digitally, it is possible to create complex, interactive content. It is easy to access a pixel in an image, or a group of pixels in a region or even a section of a sound track. Different digital operations can be applied to each, such as to enhance the image quality of a region, or to remove noise in a sound track.
- ▶ Stored digital signals do not degrade over time or distance as analog signals do. One of the most common artifacts of broadcast VHS video is ghosting, as stored VHS tapes lose their image quality by repeated usage and degradation of the medium over time.

# ANALOG AND DIGITAL SIGNALS

## Periodic signals



## Nonperiodic signals

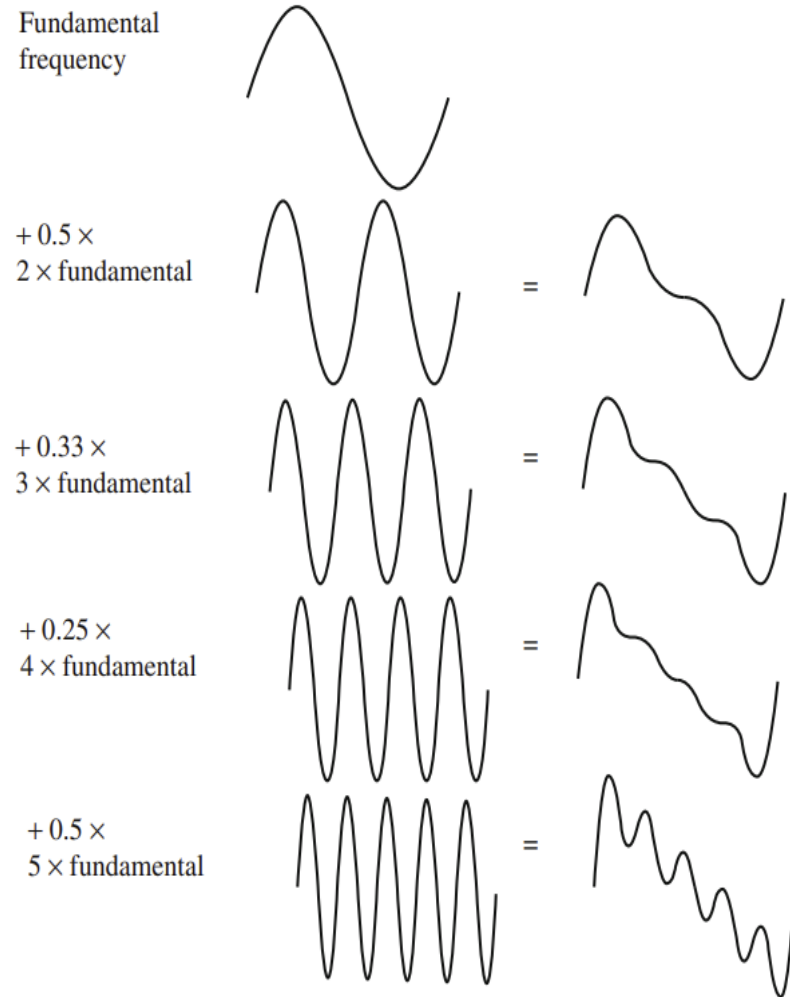


Periodic analog signals can be classified as simple or composite. A simple periodic analog signal, a sine wave, cannot be decomposed into simpler signals. A composite periodic analog signal is composed of multiple sine waves.

# Advantages of digital signals over analog

- ▶ Digital data can be efficiently compressed and transmitted across digital networks. This includes active and live distribution models, such as digital cable, video on demand, and passive distribution schemes, such as video on a DVD.
- ▶ It is easy to store digital data on magnetic media such as hard drives, or solid state memory devices, such as flash drives, memory cards. This is because the representation of digital data, whether audio, image, or video, is a set of binary values, regardless of data type.
- ▶ Digital data is preferred because it offers better quality and higher fidelity, compressed, distributed, stored, and retrieved relatively easily.

# Building complex signal

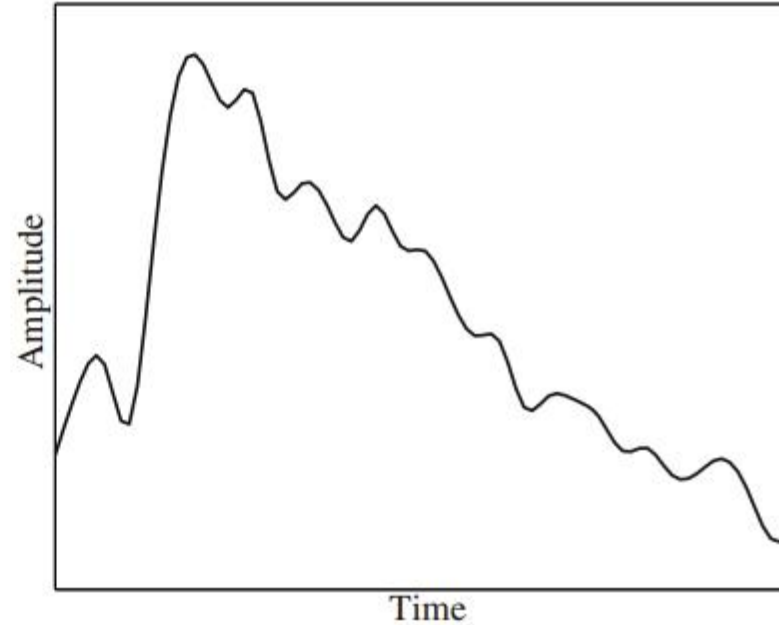


**Fig.6.1** Building up a complex signal by superposing sinusoids

sampling—using measurements only at evenly spaced time intervals—is simply called sampling, and the rate at which it is performed is called the sampling frequency

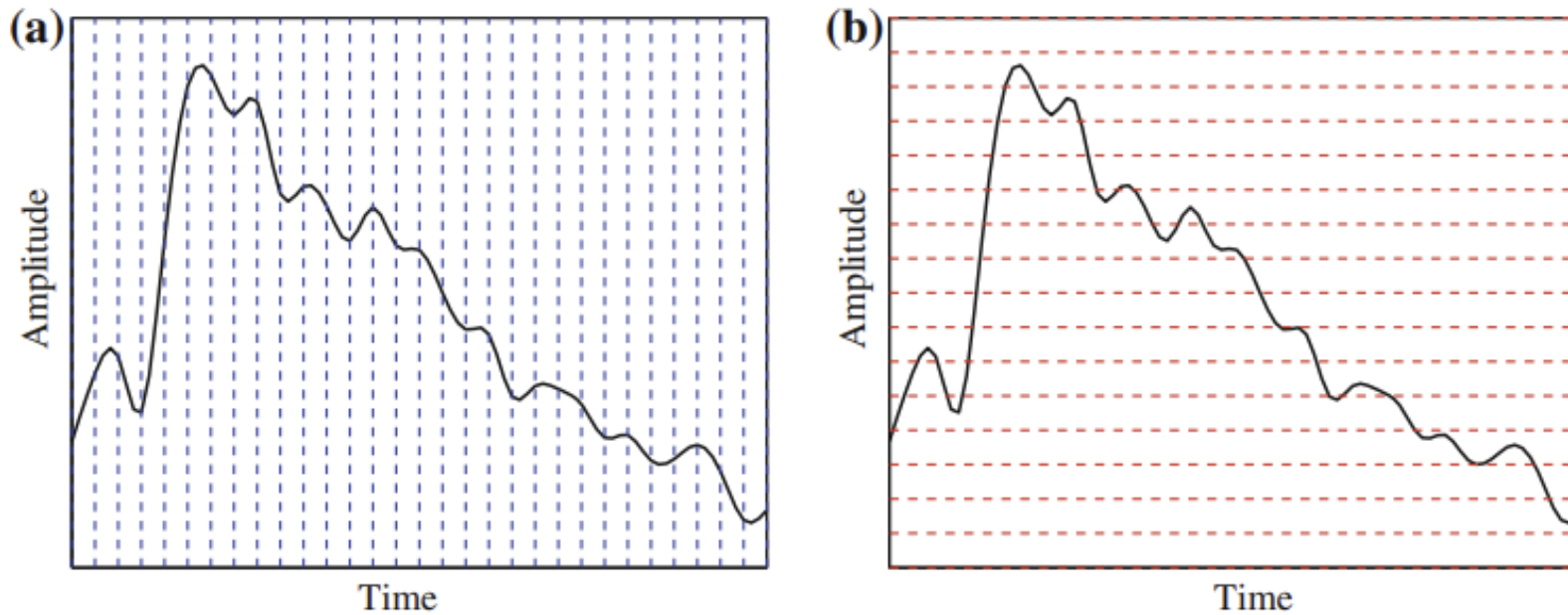
# Digitization process

**Fig. 6.2** An analog signal:  
continuous measurement of  
pressure wave



For audio, typical sampling rates are from 8 kHz (8,000 samples per second) to 48 kHz. The human ear can hear from about 20 Hz (a very deep rumble) to as much as 20 kHz; above this level, we enter the range of ultrasound. The human voice can reach approximately 4 kHz

# Sampling vs Quantization



**Fig. 6.3** Sampling and quantization: **a** sampling the analog signal in the time dimension; **b** quantization is sampling the analog signal in the amplitude dimension

# ANALOG-TO-DIGITAL CONVERSION

- ▶ The conversion of signals from analog to digital occurs via two main processes: **sampling** and **quantization**.
- ▶ The reverse process of converting digital signals to analog is known as **interpolation**.
- ▶ One of the most desirable properties in the analog to digital conversion is to ensure that no artifacts are created in the digital data. That way, when the signal is converted back to the analog domain, it will look the same as the original analog signal.

# ANALOG-TO-DIGITAL CONVERSION

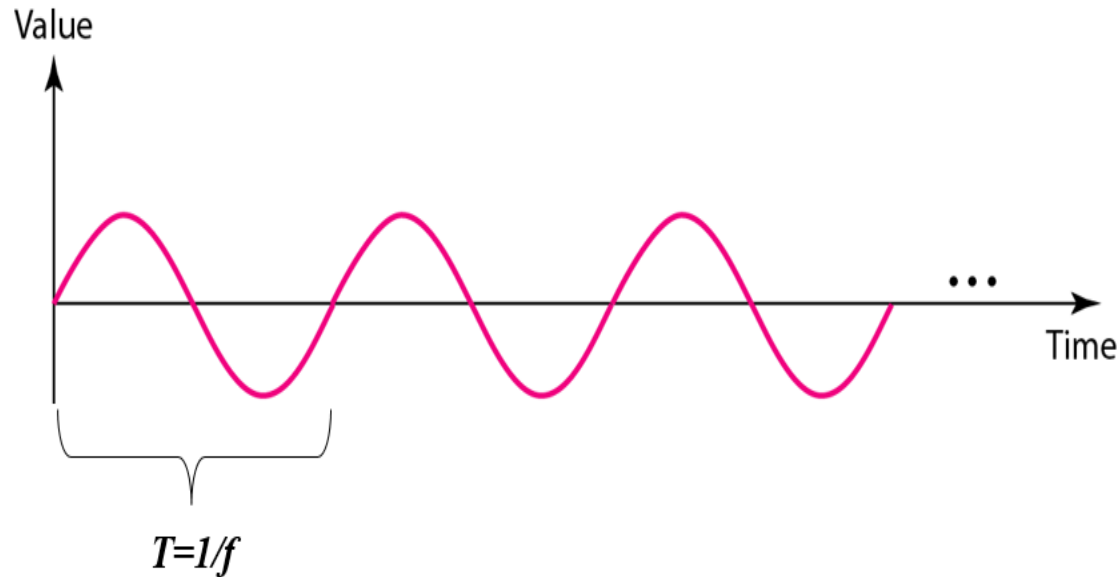


Figure: A sine wave

## Peak amplitude (A)

maximum strength of signal  
volts

## Frequency (f)

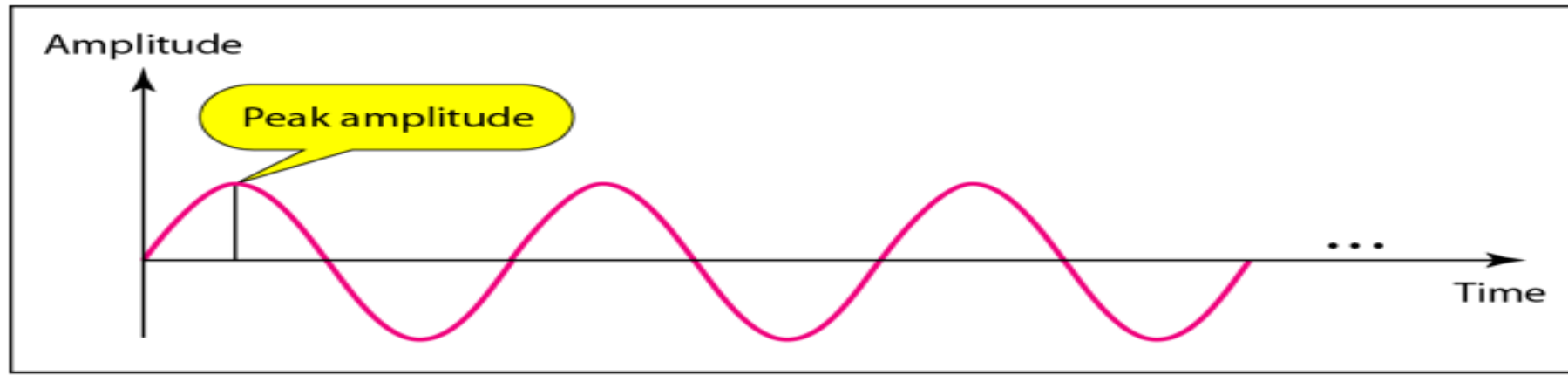
rate of change of signal  
Hertz (Hz) or cycles per second  
period = time for one repetition (T)  
 $T = 1/f$

## Phase ( $\phi$ )

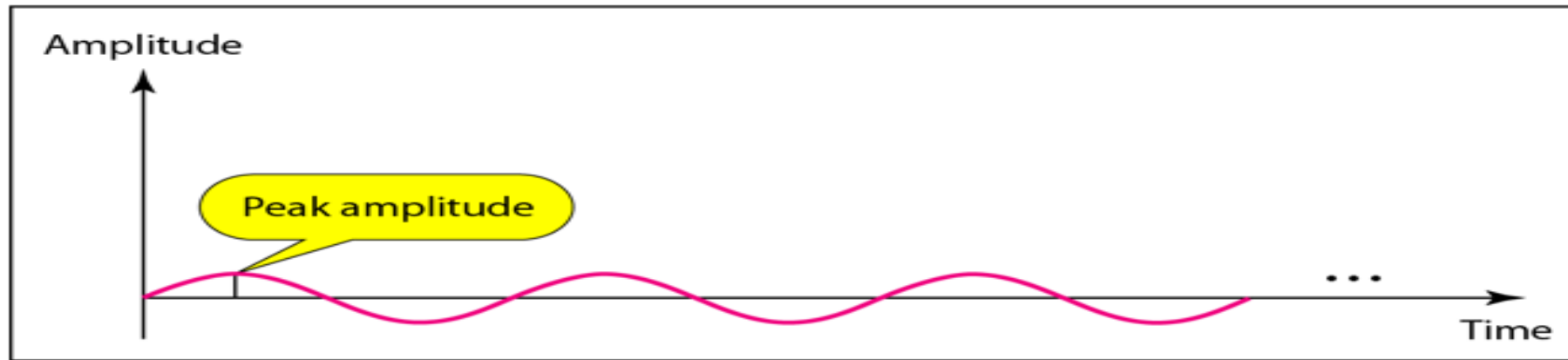
relative position in time



# ANALOG-TO-DIGITAL CONVERSION



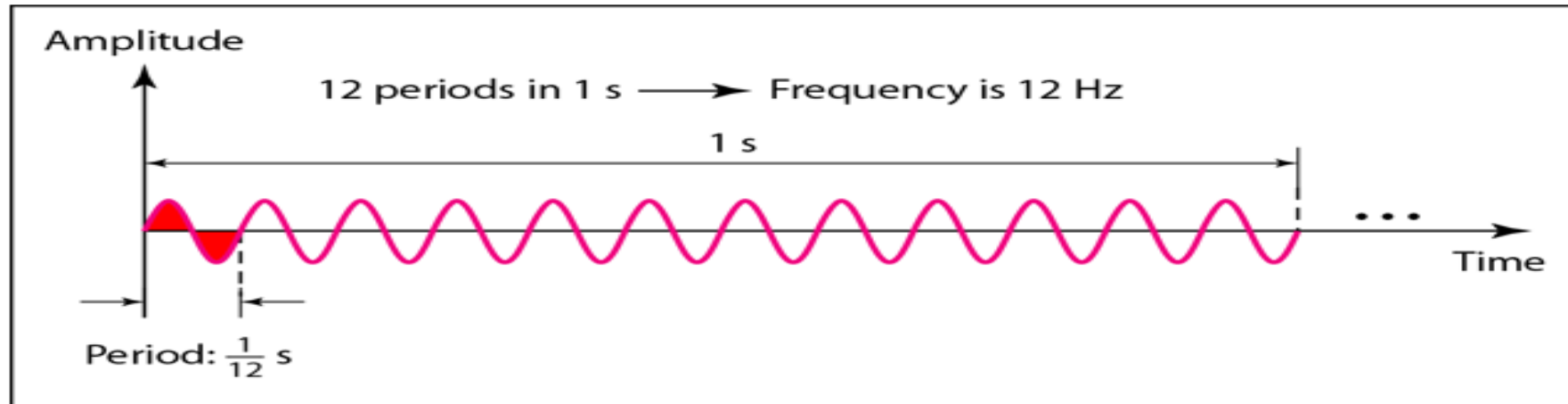
a. A signal with high peak amplitude



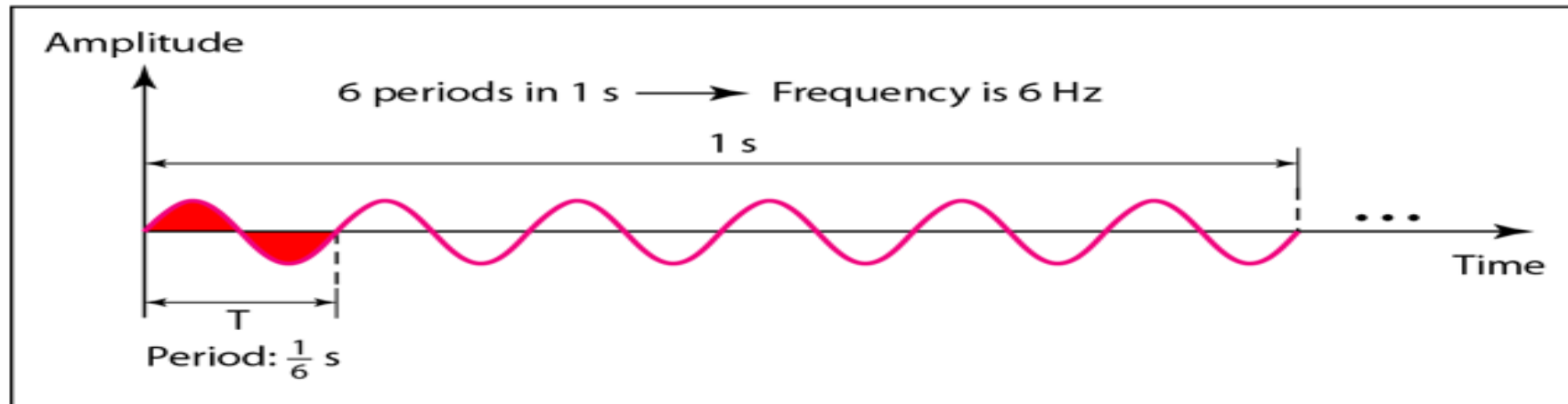
b. A signal with low peak amplitude

Figure: Two signals with the same phase and frequency, but different amplitudes

# ANALOG-TO-DIGITAL CONVERSION



a. A signal with a frequency of 12 Hz

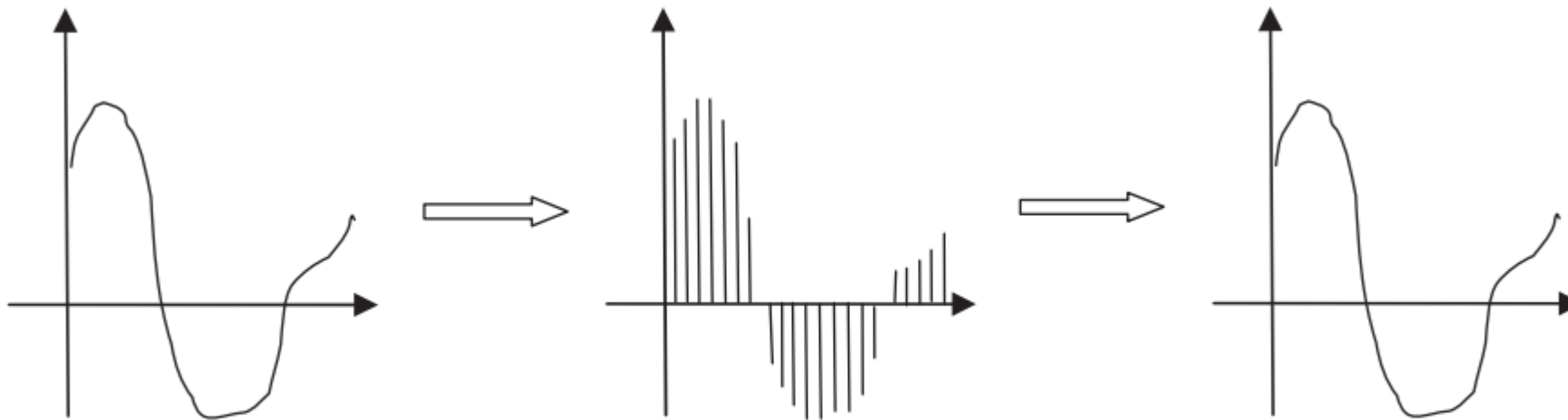


b. A signal with a frequency of 6 Hz

Figure: Two signals with the same amplitude and phase, but different frequencies

# ANALOG-TO-DIGITAL CONVERSION

- ▶ Sampling
- ▶ Assume that we start with a one-dimensional analog signal in the time  $t$  domain, with an amplitude given by  $x(t)$ . The sampled signal is given by  $x_s(n) = x(nT)$ , where  $T$  is the sampling period and  $f = 1/T$  is the sampling frequency.



**Figure 2-2 Analog-to-digital conversion and the corresponding interpolation from the digital-to-analog domain**

# ANALOG-TO-DIGITAL CONVERSION

- ▶ For commonly used signals, sampling is done across one dimension (time, for sound signals), two dimensions (spatial  $x$  and  $y$ , for images), or three dimensions ( $x$ ,  $y$ , time for video, or  $x$ ,  $y$ ,  $z$  for sampling three-dimensional ranges).
- ▶ Practical sampling involves averaging, either in time or space.

# ANALOG-TO-DIGITAL CONVERSION

- ▶ Frequency and period are the inverse of each other.

$$f = \frac{1}{T} \quad \text{and} \quad T = \frac{1}{f}$$

- Units of period and frequency

| <i>Unit</i>             | <i>Equivalent</i> | <i>Unit</i>     | <i>Equivalent</i> |
|-------------------------|-------------------|-----------------|-------------------|
| Seconds (s)             | 1 s               | Hertz (Hz)      | 1 Hz              |
| Milliseconds (ms)       | $10^{-3}$ s       | Kilohertz (kHz) | $10^3$ Hz         |
| Microseconds ( $\mu$ s) | $10^{-6}$ s       | Megahertz (MHz) | $10^6$ Hz         |
| Nanoseconds (ns)        | $10^{-9}$ s       | Gigahertz (GHz) | $10^9$ Hz         |
| Picoseconds (ps)        | $10^{-12}$ s      | Terahertz (THz) | $10^{12}$ Hz      |

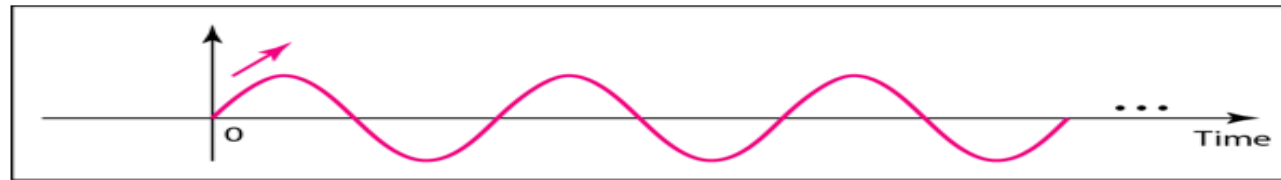
# ANALOG-TO-DIGITAL CONVERSION

- ▶ Frequency is the rate of change with respect to time.
- ▶ Change in a short span of time means high frequency
- ▶ Change over a long span of time means low frequency.

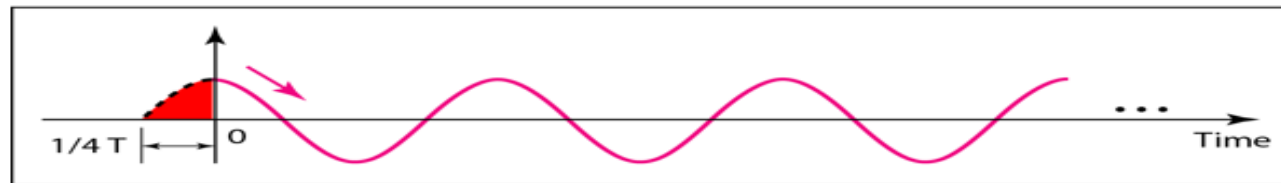
# ANALOG-TO-DIGITAL CONVERSION

- ▶ If a signal does not change at all, its frequency is zero.
- ▶ If a signal changes instantaneously, its frequency is infinite.
- ▶ Phase describes the position of the waveform relative to time 0

# ANALOG-TO-DIGITAL CONVERSION

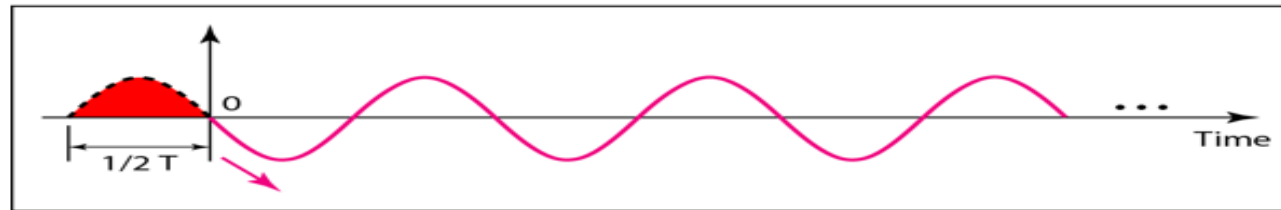


a. 0 degrees



b. 90 degrees

$\pi/2 \text{ rad}$



c. 180 degrees

$\pi \text{ rad}$

Phase unit: degree ( $360^\circ$ ) or radians ( $2\pi \text{ rad}$ )

Figure: Three sine waves with the same amplitude and frequency, but different phases



# ANALOG-TO-DIGITAL CONVERSION

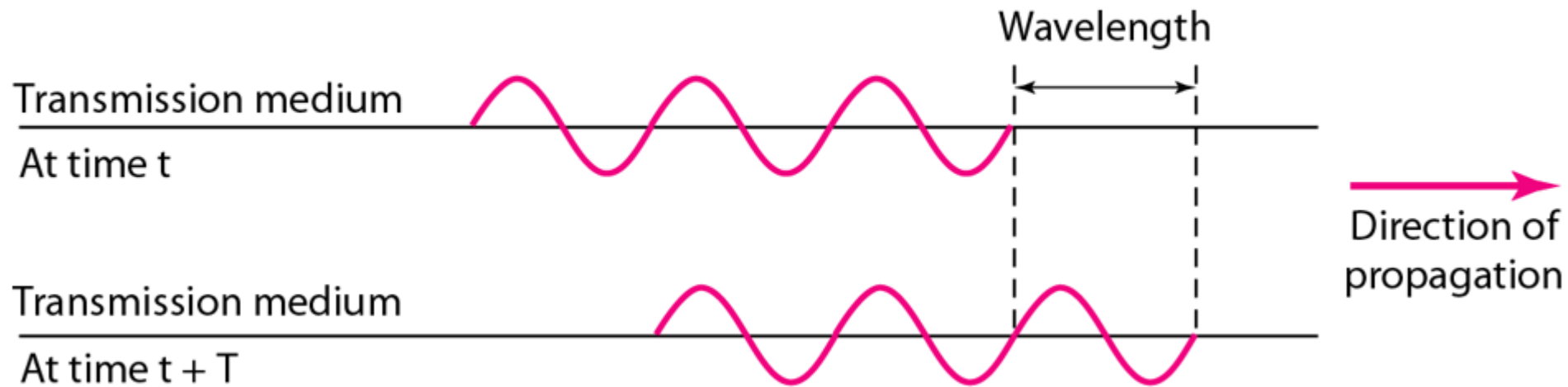
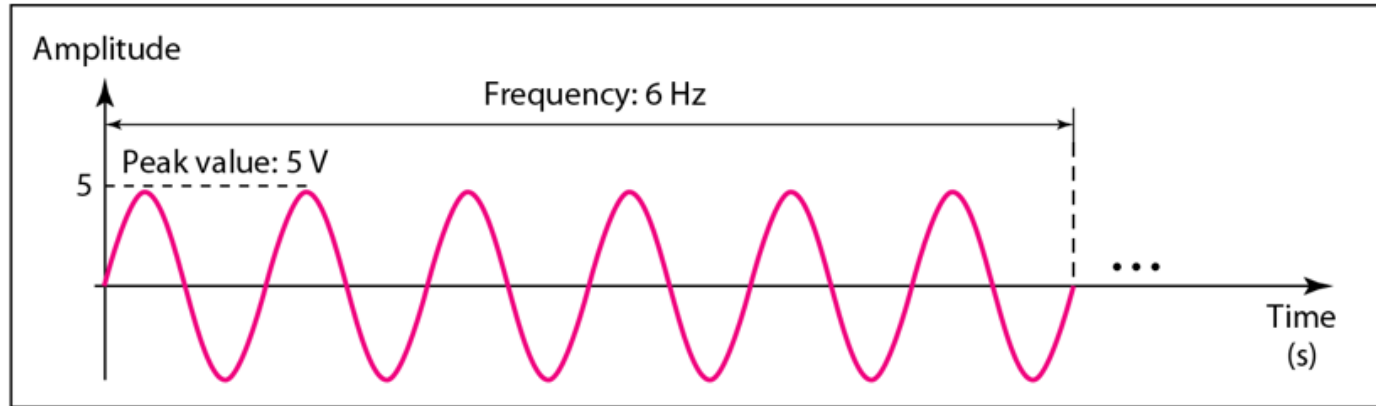


Figure: Wavelength and period

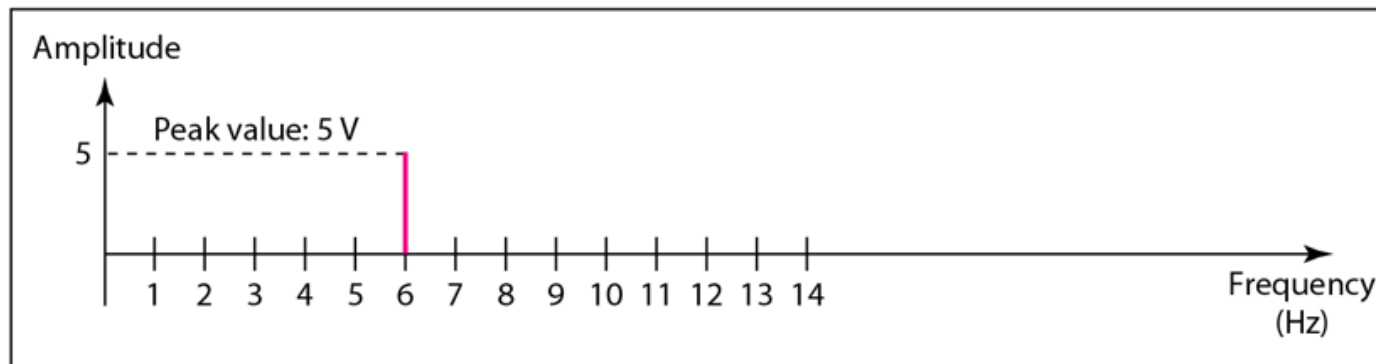
# ANALOG-TO-DIGITAL CONVERSION

- ▶ Wavelength ( $\lambda$ ) is distance occupied by one cycle
  - ▶ Another word, distance is a simple signal travels in one period.
- ▶ between two points of corresponding phase in two consecutive cycles
- ▶ assuming signal velocity  $v$  have  $\lambda = vT$
- ▶  $\lambda$  standard unit: meter (m)

# ANALOG-TO-DIGITAL CONVERSION



a. A sine wave in the time domain (peak value: 5 V, frequency: 6 Hz)



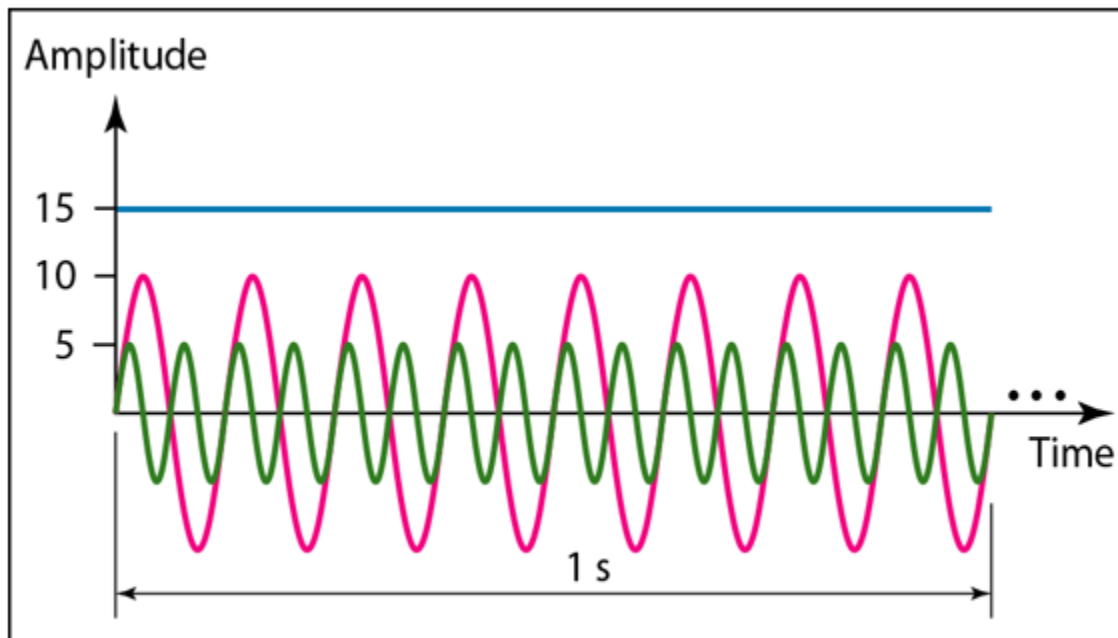
b. The same sine wave in the frequency domain (peak value: 5 V, frequency: 6 Hz)

Figure: The time-domain and frequency-domain plots of a sine wave

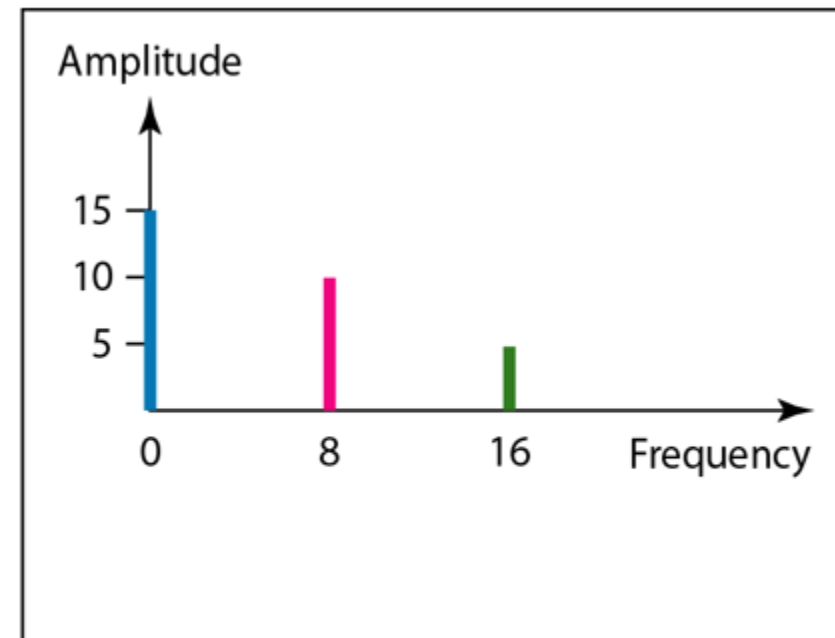
# ANALOG-TO-DIGITAL CONVERSION

- ▶ A complete sine wave in the time domain can be represented by one single spike in the frequency domain.

The frequency domain is more compact and useful when we are dealing with more than one sine wave. Figure shows three sine waves, each with different amplitude and frequency. All can be represented by three spikes in the frequency domain.



a. Time-domain representation of three sine waves with frequencies 0, 8, and 16



b. Frequency-domain representation of the same three signals

# ANALOG-TO-DIGITAL CONVERSION

- ▶ 2.2 Quantization
- ▶ Quantization deals with encoding the signal value at every sampled location with a predefined precision, defined by a number of levels.
- ▶ The entire range  $R$  of the signal is represented by a finite number of bits  $b$ . Formally,

$$x_q(n) = Q[x_s(n)], \text{ where } Q \text{ is the rounding function.}$$

$Q$  represents a rounding function that maps the continuous value  $x_s(n)$  to the nearest digital value using  $b$  bits. Utilizing  $b$  bits corresponds to  $N = 2^b$  levels, thus

having a quantization step  $\Delta = R/2^b$ .

# ANALOG-TO-DIGITAL CONVERSION

## 2.2 Quantization

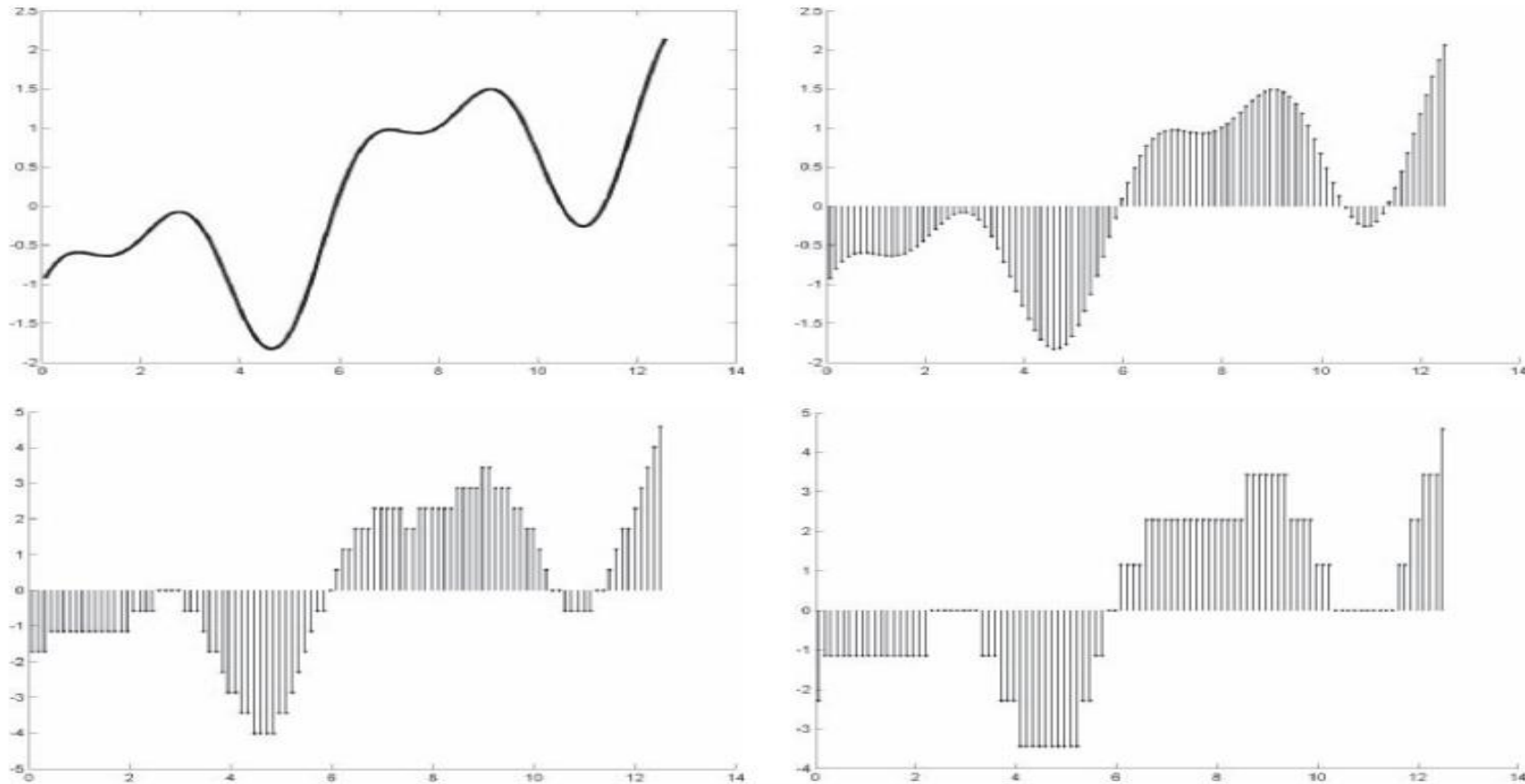


Figure 2-3 Original analog signal (upper left) is shown sampled and quantized at different quantization levels. For quantization, 8 bits (256 levels), 4 bits (16 levels), and 3 bits (8 levels) were used to produce the digital signals on the top right, bottom left, and bottom right, respectively.

# ANALOG-TO-DIGITAL CONVERSION

- ▶ Figure 2-3 shows an analog signal, which is sampled at a common frequency, but quantized using different number of bits, 4 or 2.
- ▶ Because each sample is represented by a finite number of bits, the quantized value will differ from the actual signal value, thus always introducing an error.
- ▶ The question, then, is how many bits should be used to represent each sample? Is this number the same for all signals?
- ▶ This actually depends on the type of signal and what its intended use is. **Audio signals, which represent music, must be quantized on 16 bits, whereas speech only requires 8 bits.**
- ▶ the output range of the signal is divided into fixed and uniformly separated intervals depending on the number of bits used. This works well when all the values in the range of the signal are equally likely and, thus, the quantization error is equally distributed

# ANALOG-TO-DIGITAL CONVERSION



6 bits



5 bits



4 bits



3 bits



2 bits



1 bit

Figure 2-4 Examples of quantization; initial image had 8 bits per pixel, which is shown quantized from 6 bits down to 1 bit per pixel



# ANALOG-TO-DIGITAL CONVERSION

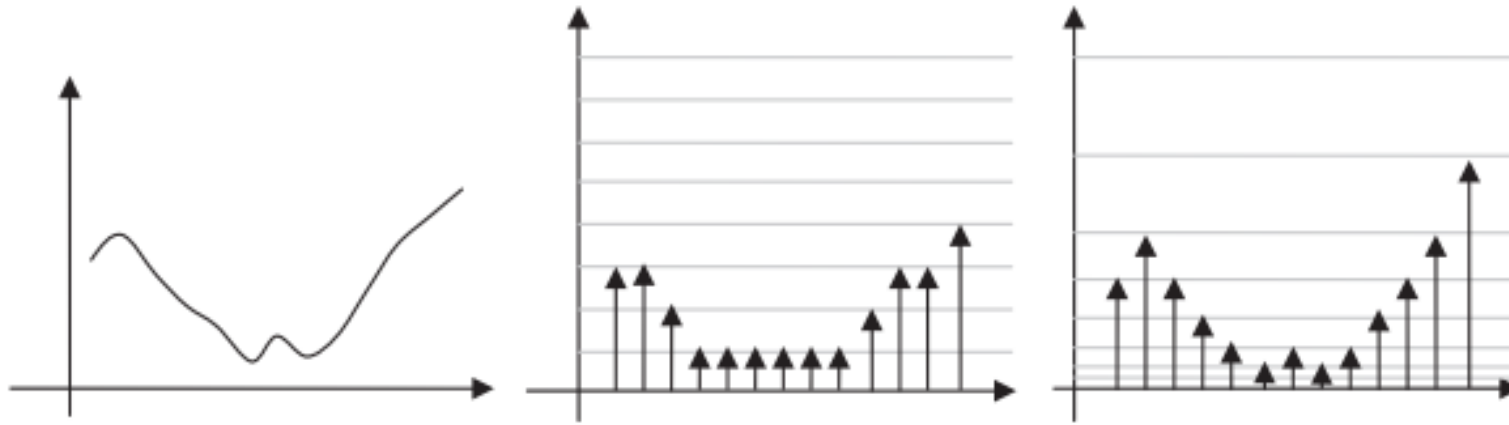


Figure 2-5 Nonlinear quantization scales. The left signal shows the original analog signal. The corresponding digitized signal using linear quantization is shown in the center. The right signal is obtained by a logarithmically quantized interval scale.

# ANALOG-TO-DIGITAL CONVERSION

- ▶ 2.3 Bit Rate
- ▶ The number of bits being produced per second. Bit rate is of critical importance when it comes to storing a digital signal, or transmitting it across networks, which might have high, low, or even varying bandwidths. Bit rate, which is measured in terms of bits per second, consists of the following:

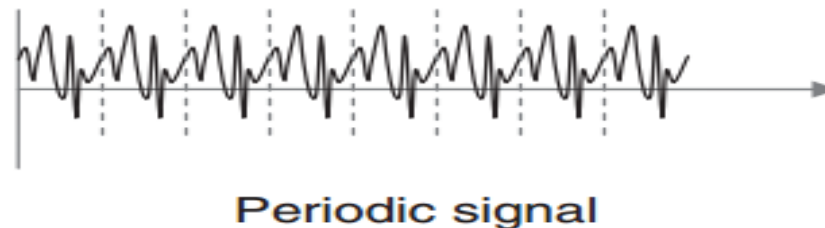
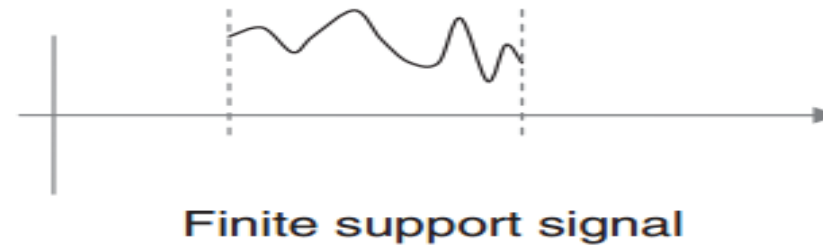
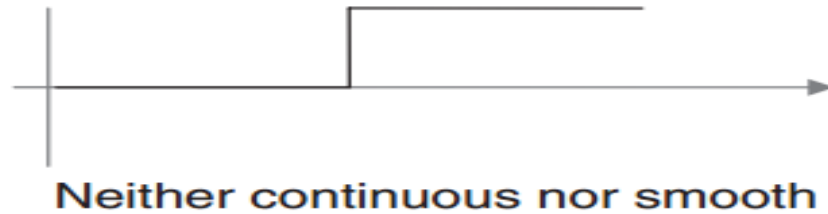
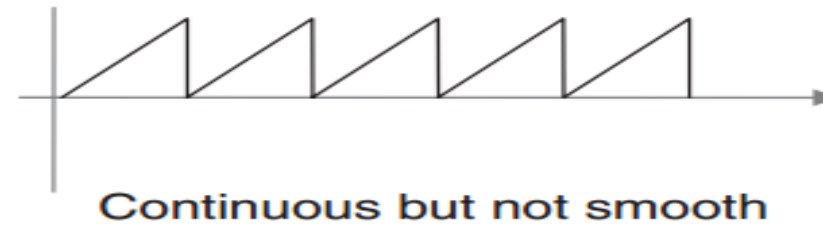
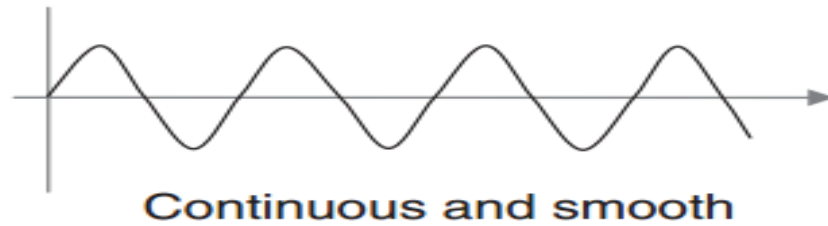
$$\begin{aligned} \text{Bit rate} &= \frac{\text{Bits}}{\text{Second}} = \left( \frac{\text{Samples produced}}{\text{Second}} \right) \times \left( \frac{\text{Bits}}{\text{Sample}} \right) \\ &= \text{Sampling rate} \times \text{Quantization bits per sample} \end{aligned}$$

# ANALOG-TO-DIGITAL CONVERSION

| Signal                 | Sampling rate                 | Quantization                    | Bit rate                             |
|------------------------|-------------------------------|---------------------------------|--------------------------------------|
| Speech                 | 8 KHz                         | 8 bits per sample               | 64 Kbps                              |
| Audio CD               | 44.1 KHz                      | 16 bits per sample              | 706 Kbps (mono)<br>1.4 Mbps (stereo) |
| Teleconferencing       | 16 KHz                        | 16 bits per sample              | 256 Kbps                             |
| AM Radio               | 11 KHz                        | 8 bits per sample               | 88 Kbps                              |
| FM Radio               | 22 KHz                        | 16 bits per sample              | 352 Kbps (mono)<br>704 Kbps (stereo) |
| NTSC TV<br>image frame | Width – 486<br>Height – 720   | 16 bits per sample              | 5.6 Mbits per frame                  |
| HDTV (1080i)           | Width – 1920<br>Height – 1080 | 12 bits per pixel<br>on average | 24.88 Mbits per<br>frame             |

Figure 2-6 Table giving the sampling rate, quantization factor, and bit rates produced for typical signals

# SIGNALS AND SYSTEMS



- Figure 2-7 Sample signals with different kinds of properties—smooth, unsmooth, continuous, discontinuous, finite support, and periodic. Normally, signals are composed of a combination of one or more of these properties.

# SIGNALS AND SYSTEMS

- Continuous and smooth—Such as a sinusoid.
- Continuous but not smooth—Such as a saw tooth.
- Neither smooth nor continuous—For example, a step edge.
- Symmetric—Which can be further described either as odd ( $y = \sin(x)$ ) or even ( $y = \cos(x)$ ). Note that any signal can be decomposed into the sum of an odd and even part.
- Finite support signals—Signals that are defined for a finite interval and zero outside of that interval.
- Periodic signal—A signal that repeats itself over a time period. For a periodic signal  $f(x)$ , the period is defined to be  $T$  if  $f(x + T) = f(x)$ . For any function  $f(t)$ , we can define a periodic version of  $f(t)$ ,  $g(t) = \sum_k f(t - kT)$ .

# Linear Time Invariant Systems

- ▶ Any operation that transforms a signal is called a system. Understanding linear time invariant systems is necessary to gain insight into the fundamental results that now characterize the process by which any practical system performs sampling and digitization.

Let a system transform an input signal  $x(t)$  into an output  $y(t)$ . We call this system a linear if the output and input obey the following:

$$\begin{aligned} \text{If} \quad & x(t) = c_1 x_1(t) + c_2 x_2(t) \\ \text{then} \quad & y(t) = c_1 y_1(t) + c_2 y_2(t) \\ \text{where } & y_k(t) \text{ is the sole output resulting from } x_k(t) \end{aligned}$$

# Linear Time Invariant Systems

- ▶ Another important operation that is used to process signals in an LTI system is convolution. The convolution of two signals  $f$  and  $g$  is mathematically represented by  $f * g$ . It is the result of taking the integral of the first signal multiplied with the other signal reversed and shifted.

$$\begin{aligned}(f * g) &= \int_{-\infty}^{\infty} f(\tau) \cdot g(t - \tau) d\tau \\ &= \int_{-\infty}^{\infty} f(t - \tau) \cdot g(\tau) d\tau\end{aligned}$$

# Linear Time Invariant Systems

- ▶ Any LTI system is fully characterized by a specific function, which is called the impulse response of the system. The output of the system is the convolution of the input with the system's impulse response. This analysis is termed as the time Domain point of view of the system.
- ▶ Alternatively, we can also express this result in the frequency domain by defining the system's transfer function. The transfer function is the Fourier transform of the system's impulse response



# Linear Time Invariant Systems

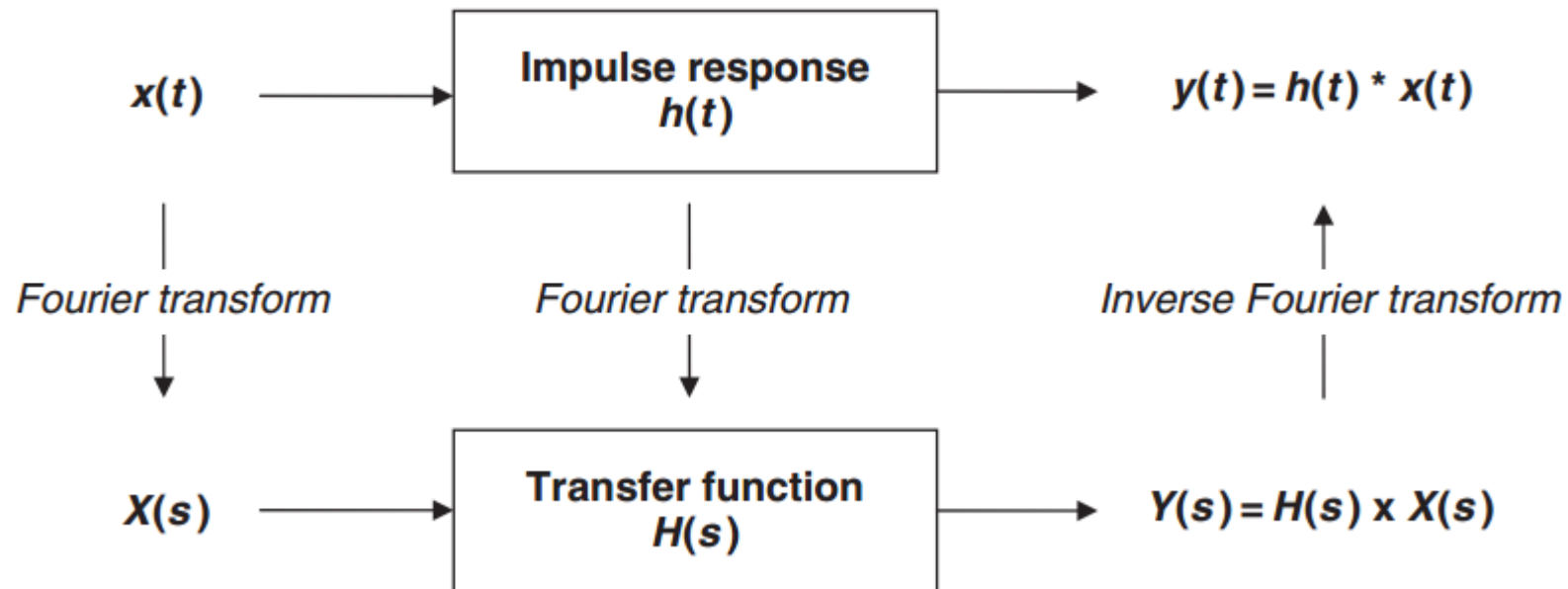


Figure 2-8 Relationship between the impulse response function in the time domain and the transfer function in the frequency domain

## 3.3 Useful Signals

- Specific signals—for example, the Delta function, box function, step function—play an important role in many signaling-processing operations, such as sampling, filtering, and convolution

The Delta function is a mathematical construct introduced by the theoretical physicist Paul Dirac and, hence, is also known as the Dirac Delta function. Informally, it is a sharp peak bounding a unit area:  $\delta(x)$  is zero everywhere except at  $x = 0$  where it becomes infinite, and its integral is 1

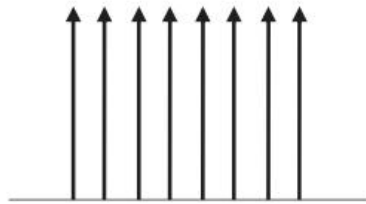
# The Fourier Transform

- ▶ The Fourier transform is due to Joseph Fourier (1768-1830), who published his initial results in 1807. **Fourier proposed to represent any periodic, continuous signal as a sum of individual complex sinusoids** (a Fourier series expansion).
- ▶ In other words every periodic, continuous signal can be expressed as a weighted combination of sinusoid (sine and cosine) waves. The weights that are used to combine the sinusoids are called the Fourier series coefficients, or spectral components.

# The Fourier Transform

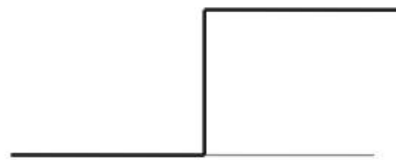


Delta function

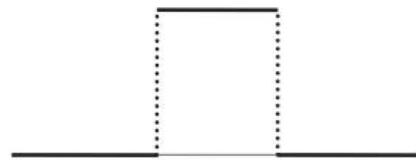


Comb  
function

$$\Delta_T(t) \stackrel{\text{def}}{=} \sum_{k=-\infty}^{\infty} \delta(t - kT)$$

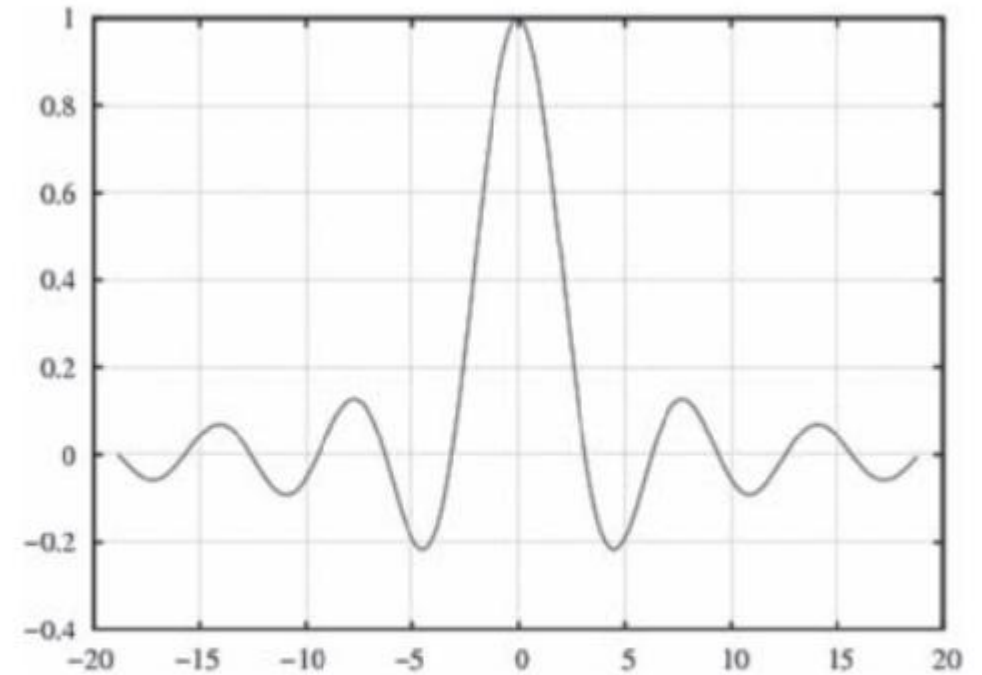


Step function  $H(x) = \int_{-\infty}^x \delta(t)dt$



Box function

$$\text{rect}(t) = \Pi(t) = \begin{cases} 0 & \text{if } |t| > \frac{1}{2} \\ \frac{1}{2} & \text{if } |t| = \frac{1}{2} \\ 1 & \text{if } |t| < \frac{1}{2} \end{cases}$$



Sinc function  $\text{Sa}(x) = \text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$

Figure 2-9 Useful functions in signal-processing theory: Delta function (top left), comb function (top right), step function (middle left), box function (middle right), and sinc function (right)

# Some sample signals

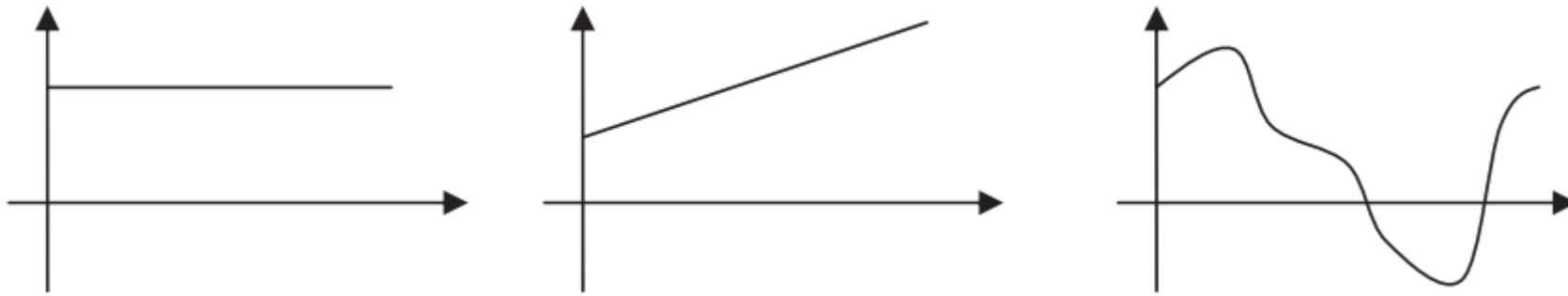


Figure 2-10 Examples of simple one-dimensional functions. Different numbers of samples are required to digitize them

Some sample signals are shown in Figure 2-10

# SAMPLING THEOREM AND ALIASING

- ▶ If these signals are to be digitized and reproduced back in the analog domain, the number of samples required to ensure that both starting and ending analog signals are the same is clearly different. For instance, in the first case where the signal is not changing (and, hence, has zero frequency), one single sample will suffice for this purpose.
- ▶ Two samples are required for the second signal and many more for the third one. As we go from left to right, the frequency content in the signals increases, and, therefore, the number of samples needed during the digitization process also goes up

# SAMPLING THEOREM AND ALIASING

- ▶ The relationship between signals and sampling rate was established during the late 1920s by Henry Nyquist and later formalized by Claude Shannon in 1950.
- ▶ This relationship states that the signal has to be sampled using a sampling frequency that is greater than **twice the maximal frequency occurring in the signal**

# SAMPLING THEOREM AND ALIASING

it can more formally be stated as follows:

1. A bandlimited signal  $f(t)$  with max frequency  $\omega F$  is fully determined from its samples  $f(nT)$  if  $2\pi/T > 2\omega F$
2. The continuous signal can then be reconstructed from its samples  $f(nT)$  by convolution with the filter

$$r(t) = \text{sinc}(\omega F(t - nT)/2\pi)$$

For example, if the signal has a maximal frequency of 10 kHz, it should be sampled at a frequency greater than 20 kHz. Here, 20 kHz is known as the Nyquist sampling frequency for that signal.



# SAMPLING THEOREM AND ALIASING

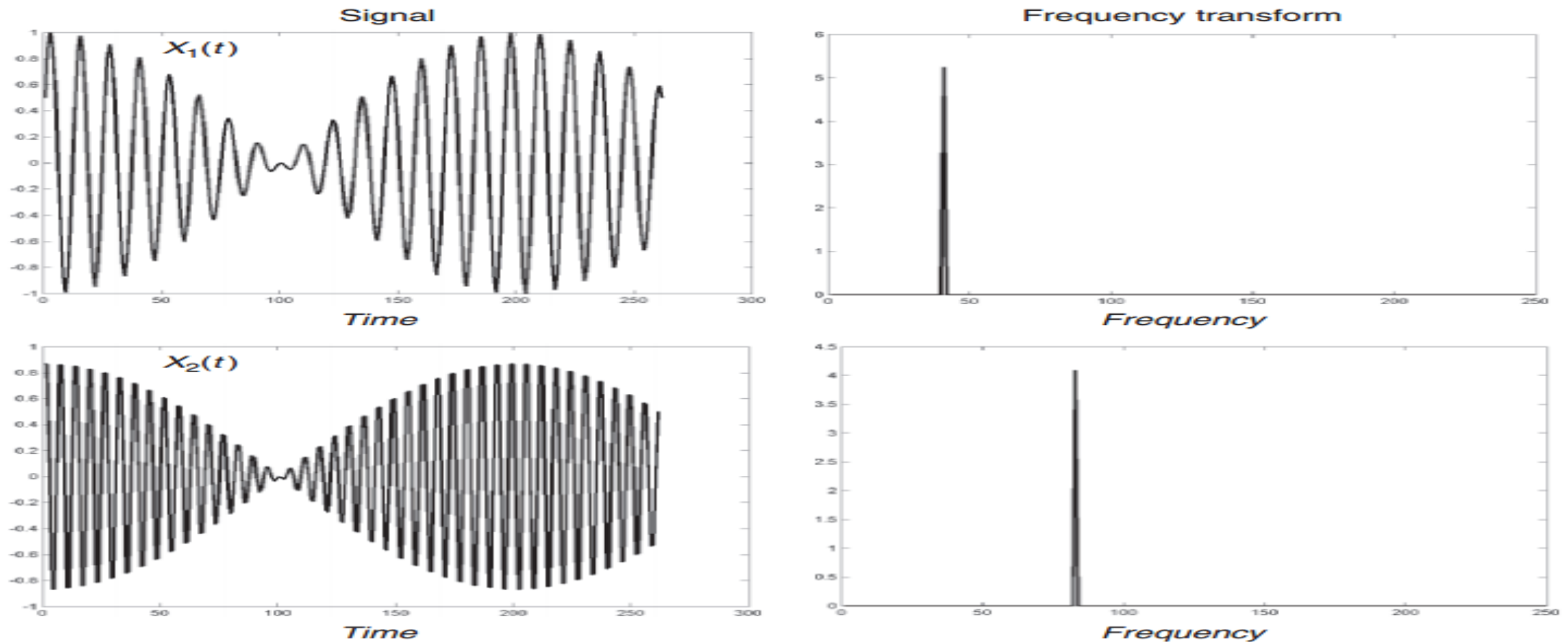


Figure 2-11 Two signals  $x_1(t)$  and  $x_2(t)$  having different frequencies are shown on the left and their Fourier transforms on the right. The transforms show the amount of energy at all frequencies. Both input signals have just one frequency as their Fourier transform shows, but the frequency in  $x_1(t)$  is lower than that in  $x_2(t)$

# SAMPLING THEOREM AND ALIASING

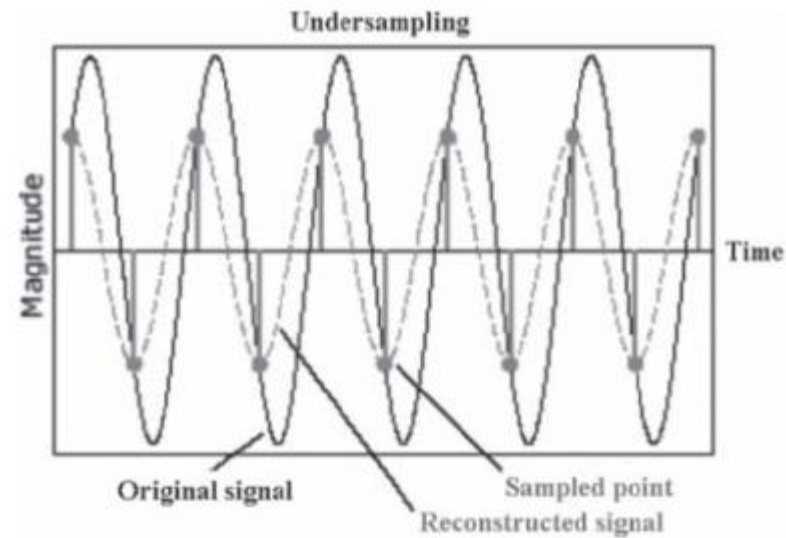
- ▶ What happens if your sampling frequency is higher than your Nyquist frequency? The answer is: nothing special.
- ▶ When it comes to reproducing your analog signal, it is guaranteed to have all the necessary frequencies and, hence, the same signal is reproduced.
- ▶ However, by performing such over sampling, you have generated more digital data than necessary, which increases your storage and transmission overhead.

# SAMPLING THEOREM AND ALIASING

- ▶ What happens if your sampling frequency is lower than your Nyquist frequency?
- ▶ In this case, you have a problem because all the frequency content is not well captured during the digitization process. As a result, when the digital signal is heard/viewed or converted back into analog form, it does not correspond to the initial starting analog signal.
- ▶ Aliasing is the term used to describe loss of information during digitization

# SAMPLING THEOREM AND ALIASING

- ▶ Aliasing in Spatial Domains
- ▶ Aliasing effects in the spatial domain are seen in all dimensions. Figure 2-12 illustrates examples in one and two dimensions.



# SAMPLING THEOREM AND ALIASING



Figure 2-12 Aliasing examples in the spatial domain. The top figure shows an example in one dimension, where original signal is shown along with sampled points and the reconstructed signal. The bottom set of figures show a 2D input image signal. The top left shows the original signal and the remaining three show examples of the signal reconstruction at different sampling resolutions.

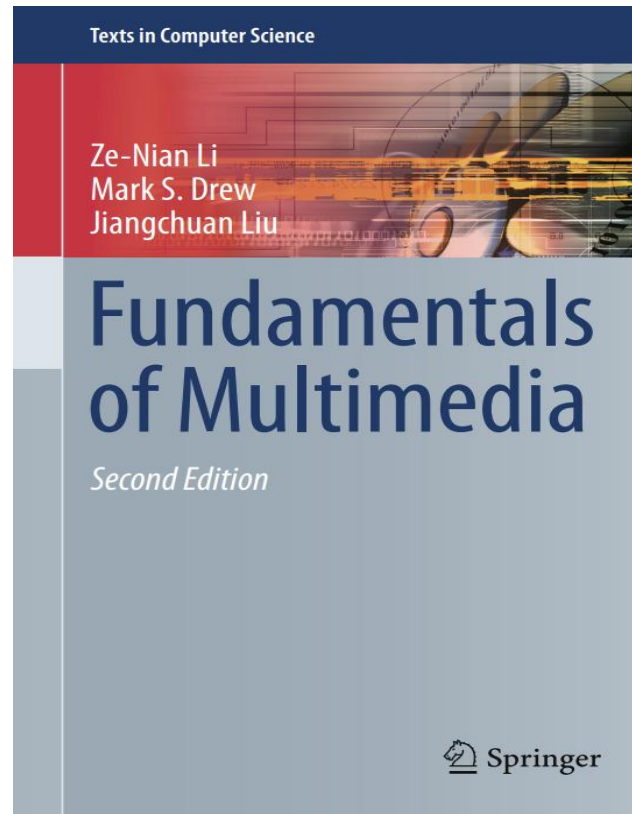
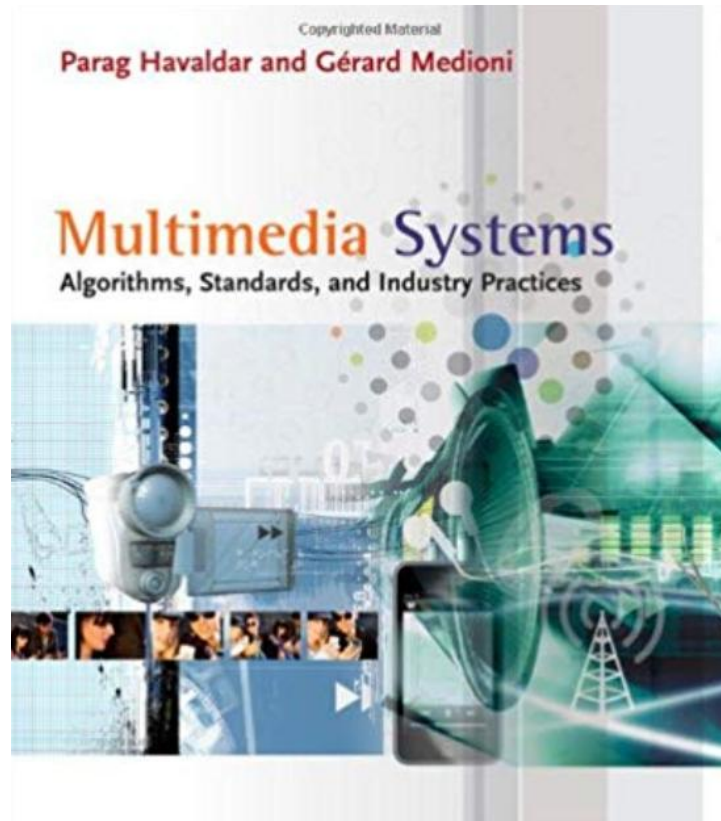
# SAMPLING THEOREM AND ALIASING

- ▶ In the two-dimensional example, the first image shows the original image consists of 750 samples in the x and 620 samples in the y direction. The succeeding figures show what the reconstructed signal looks like by reducing the sampling resolution in both directions.
- ▶ You can see that, as the number of spatial samples decreases, not all of the original frequencies are properly captured.

# SAMPLING THEOREM AND ALIASING

- ▶ Aliasing in the Temporal Domain
- ▶ Examples of temporal aliasing can be seen in western movies, by observing the motion of stagecoach wheels. As the stagecoach starts to move, we observe its wheels rotating in the expected forward direction. As the stagecoach speeds up, we see that the wheels appear to rotate in the opposite direction to the initial one.
- ▶ This occurs because the movie camera has under sampled the wheels' motion

# Reference





# Question



Thank you

